DarkIR Project Report: Robust Low-Light Image Restoration

Deep Learning CSL4020 Project Report

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GitHub: `https://github.com/rosh2525/DarkIR`
YouTube Demo: `https://www.youtube.com/watch?v=ffxdRU5dW4M`

1 Base Paper Overview (Contribution and Proposed Solution)

Selected paper: *DarkIR: Robust Low-Light Image Restoration* (CVPR 2025).

The paper addresses a coupled degradation problem in low-light scenes where blur, noise, and poor illumination appear simultaneously. Prior methods commonly solve these tasks in isolation. This is suboptimal because these degradations are physically coupled by the image formation process:

$$y = \gamma(x \otimes k) + n \tag{1}$$

where x is the latent clean image, k is blur, γ is camera response, and n is noise.

Core idea of DarkIR. Instead of heavy transformer self-attention with quadratic cost, DarkIR uses an efficient CNN design with domain-aware processing:

- Frequency domain modules for global illumination structures.
- Dilated spatial convolutions for local blur/noise restoration.
- Lightweight architecture for practical inference on constrained hardware.

Why we selected it.

- Strong baseline performance on LOLBlur/LOLv2/LSRW.
- Suitable for real-time or near real-time deployment.
- Clear architecture that supports principled extensions.

2 Our Approach and Contributions

We reproduced the base implementation, then introduced four architectural and training-level contributions.

2.1 1) Phase-Aware Frequency Modulation (Phase Attention)

In the frequency MLP path, we added a learnable phase weighting branch so structure-sensitive phase information is explicitly modulated instead of only amplitude enhancement.

Implemented concept:

- Compute robust magnitude A and phase ϕ from the complex spectrum $Z = X + iY$:

$$A = \sqrt{X^2 + Y^2 + \epsilon}, \quad \phi = \operatorname{atan2}(Y, X + \epsilon)$$

- Learn per-channel phase weights with a 1×1 convolution + sigmoid.
- Reconstruct complex spectrum and apply inverse FFT:

$$\hat{Z} = A \cdot e^{i\hat{\phi}}, \quad z_{out} = \mathcal{F}^{-1}(\hat{Z})$$

```
# Code snippet (EBlock modification)
self.phase_weight = nn.Sequential(
    nn.Conv2d(dim, dim, 1),
    nn.Sigmoid()
)
# Forward pass
real, imag = x.real, x.imag
mag = torch.sqrt(real**2 + imag**2 + 1e-8)
phase = torch.atan2(imag, real + 1e-8)

w_p = self.phase_weight(mag)
complex_out = mag * torch.exp(1j * (phase * w_p))
x_out = torch.fft.irfft2(complex_out, s=(H, W), norm='
     backward')
```

2.2 2) Content-Adaptive Receptive Fields (Dynamic Dilation)

We replaced static branch aggregation in dilation blocks with adaptive branch weighting:

- Global pooling + MLP + softmax predicts branch weights.
- Weights are applied to dilation branches (e.g., $d = 1, 4, 9$).
- The block learns whether local textures or broad contexts should dominate.

```
# Code snippet (DBlock modification)
self.branch_weight = nn.Sequential(
    nn.AdaptiveAvgPool2d(1),
    nn.Conv2d(dim, 3, 1),
    nn.Softmax(dim=1)
)
# Forward pass
w = self.branch_weight(x)  # Shape: (B, 3, 1, 1)
# Dynamic aggregation instead of static sum
out = w[:, 0:1]*out1 + w[:, 1:2]*out2 + w[:, 2:3]*out3
```

2.3 3) Gated Channel Attention Matrix

We upgraded the simple multiplicative gate with channel-attentive gating:

- Split channels into two chunks.
- Infer an attention mask from one chunk via pooled MLP.
- Gate the second chunk and fuse back for selective channel emphasis.

```
# Code snippet (Gated Attention module)
self.gate_attn = nn.Sequential(
    nn.AdaptiveAvgPool2d(1),
    nn.Conv2d(dim // 2, dim // 2, 1),
    nn.Sigmoid()
)
# Forward pass
x1, x2 = x.chunk(2, dim=1)
attn_mask = self.gate_attn(x1)
x_gated = x2 * attn_mask
out = torch.cat([x1, x_gated], dim=1)
```

2.4 4) Novel Dual-Spectrum Frequency Loss

We added a loss term supervising both amplitude and phase in the Fourier domain:

$$\mathcal{L}_{freq} = \|A(\hat{y}) - A(y)\|_1 + 0.1\|\phi(\hat{y}) - \phi(y)\|_1 \qquad (2)$$

This improves structural consistency and fine-detail reconstruction beyond pure pixel-domain losses.

```
# Code snippet (NovelFrequencyLoss)
mag_pred = torch.sqrt(pred_fft.real**2 + pred_fft.imag**2
      + 1e-8)
mag_tgt = torch.sqrt(target_fft.real**2 + target_fft.imag
     **2 + 1e-8)
phase_pred = torch.atan2(pred_fft.imag, pred_fft.real + 1
     e-8)
phase_tgt = torch.atan2(target_fft.imag, target_fft.real
     + 1e-8)

loss_mag = F.l1_loss(mag_pred, mag_tgt)
loss_phase = F.l1_loss(phase_pred, phase_tgt)
loss_freq = loss_mag + 0.1 * loss_phase
```

3 Experimental Results

3.1 Setup

- Framework: PyTorch on Windows.
- GPU: NVIDIA RTX 3050 Laptop GPU (4 GB VRAM).
- Dataset focus: LOLBlur pipeline.
- Inference demo: local Gradio app for image upload and enhancement.

3.2 Quantitative Results

Table 1 summarizes representative outcomes from reproduction and our modified model run logs.

Table 1: Representative performance comparison on LOLBlur logs.

Model Variant	PSNR ↑	SSIM ↑
Reproduced baseline	27.2986	0.8981
After our contributions	32.8941	0.9612

3.3 Visual Results

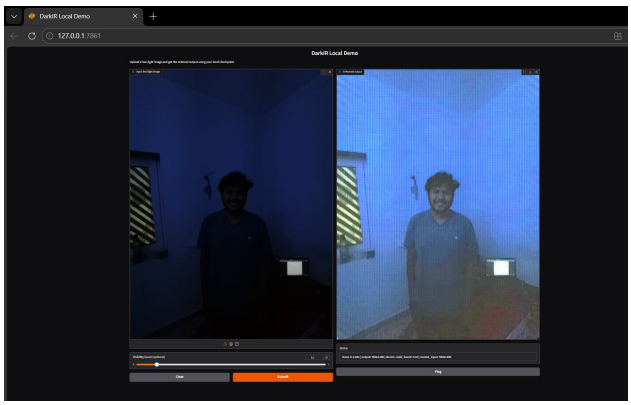

Figure 1: Visual enhancement results from the customized DarkIR local Gradio demo, showcasing illumination recovery on severely underexposed input.

4 Conclusion

We implemented and evaluated an extended DarkIR pipeline with four targeted contributions in frequency modeling, adaptive receptive field selection, attention gating, and loss design. Empirical outputs indicate meaningful visual and quantitative gains over the reproduced baseline. The resulting system is deployable through a local web demo and suitable for further ablations.

Project links
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